LEVEL UP ACADEMY

Class 10 | GSEB | Science

Ch 8 Test (GSEB pattern): Heredity

Chapter / Topic: Heredity

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory.
2. Section A: objective questions, 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 4 marks.
3. Write balanced equations and draw neat, labelled diagrams wherever needed.

SECTION A

9 marks

Choose the correct option.

Q1. Tall pea plants with violet flowers were crossed with short pea plants with white flowers. All the progeny had violet flowers, but almost half of them were short. The genetic make-up of the tall parent is: **[1]**

(a) TTWW (b) TtWW
(c) TTww (d) TtWw

Q2. Plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr). The F_1 seeds are: **[1]**

(a) all round (b) all wrinkled
(c) half round and half wrinkled (d) 3 round : 1 wrinkled

Q3. In human beings, the sex of the child is decided by: **[1]**

(a) the mother (b) the father
(c) both parents equally (d) the environment

Q4. How many pairs of chromosomes are there in human body cells? **[1]**

(a) 21 (b) 22
(c) 23 (d) 46

Q5. In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is: **[1]**

(a) 1 : 2 : 1 (b) 9 : 3 : 3 : 1
(c) 1 : 1 (d) 3 : 1

Q6. A child has the sex chromosomes XX. The child is: **[1]**

(a) a boy (b) impossible to predict
(c) either a boy or a girl (d) a girl

Q7. Genes are located on the: **[1]**

(a) ribosomes (b) cell membrane
(c) chromosomes (d) cytoplasm

Q8. In the F_2 generation of Mendel's dihybrid cross, what fraction of plants show both recessive traits (wrinkled and green)? **[1]**

(a) 1/16 (b) 3/16
(c) 1/4 (d) 9/16

Q9. When a pure tall (TT) pea plant is crossed with a pure short (tt) plant, the F_1 plants are: **[1]**

(a) all tall (b) all short
(c) half tall and half short (d) 3 tall : 1 short

SECTION B

6 marks

Q10. What are dominant and recessive traits? Give one example of each from Mendel's experiments. **[2]**

Q11. Why were all the F_1 plants tall, with no medium-height plants, when Mendel crossed pure tall and pure short pea plants? **[2]**

Q12. How is the sex of a child determined in human beings? **[2]**

SECTION C

6 marks

Q13. Mendel crossed pea plants with round, yellow seeds (RRYY) with plants having wrinkled, green seeds (rryy). **[3]**

(i) What were the F_1 seeds like? **[1]**

(ii) Write the F_2 phenotypic ratio. **[1]**

(iii) What does this cross show about the inheritance of the two traits? **[1]**

Q14. A pea plant with violet flowers (Vv) is crossed with a plant with white flowers (vv). **[3]**

(i) Write the gametes produced by each parent. **[1]**

(ii) Write the genotypes of the offspring and their ratio. **[1]**

(iii) Write the phenotypic ratio. **[1]**

SECTION D

4 marks

Read the passage and answer the questions that follow.

Q15. A student crossed pea plants with violet flowers (VV) with plants having white flowers (vv). All F_1 plants had violet flowers. When the F_1 plants were self-pollinated, out of 400 F_2 plants, about 300 had violet flowers and 100 had white flowers. **[4]**

(i) Which flower colour is dominant? **[1]**

(ii) Write the genotype of the F_1 plants. **[1]**

(iii) Out of the 300 violet-flowered F_2 plants, how many are expected to be pure-breeding (VV)? Explain. **[2]**

~ All the best! ~